

Global Proof Architecture Companion Note

Series Interlock: $SU(2)$, $SU(N)$, Classical Families, Exceptional Groups

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March 2026

Purpose. This note is a referee-facing map of how the four manuscripts in the series interlock. It adds no new claims and contains no new lemmas. Its purpose is navigational: to make explicit which components are (i) group-agnostic (shared), (ii) family-dependent, and (iii) reused verbatim across the papers, so a referee can verify the series without repeatedly re-deriving the same logic.

Series objects

- **P1:** *Yang–Mills Mass Gap for $SU(2)$ in Four Dimensions* (base case; full end-to-end template).
- **P2:** *Yang–Mills Mass Gap for $SU(N)$ in Four Dimensions* (unitary family; A1 gate via Toeplitz/total positivity).
- **P3:** *Yang–Mills Mass Gap for Orthogonal and Symplectic Groups in Four Dimensions* (Spin/Sp families; A1 gates in Appendices G–I).
- **P4:** *Yang–Mills Mass Gap for the Exceptional Groups in Four Dimensions* (G_2 , F_4 , E_6 , E_7 , E_8 ; A1 gate via exceptional-group harmonic analysis).

High-level decomposition

Every manuscript is organized into the same three functional blocks: (i) **Lane C** (continuum flowed state by dyadic RG), (ii) **Lane A** (sharp-local limit and OS axioms), and (iii) **Route 1 / Lane B** (one-shot KP entrance $\rightarrow$ clustering $\rightarrow$ OS gap intake). Only a single analytic gate is genuinely group-dependent: the one-plaquette coefficient crossover bound **A1**. Everything else is universal finite-dimensional block geometry, polymer/KP bookkeeping, or OS reconstruction technology.

Block 0: Common conventions and scope controls

All four papers fix the same interpretive scope: the constructed theory is the OS vacuum representation of the adjoint-local sharp local net generated by (flowed) curvature composites. Consequently, the claims are about strictly local gauge-invariant observables; no claim is made about sectors generated by nonlocal line/surface operators. For global-form variations (Spin vs SO, center quotients), the observable algebra descends to the effective quotient $G/\ker(\rho)$ induced by the chosen defining representation.

Block 1: Route 1 is finite-depth and modular

Route 1 has a fixed logical signature across the series:

1. **One-shot entrance:** prove that at a single entrance scale $L_{\text{ent}}(\beta)$ the exact blocked specification lies in a KP (Kotecký–Preiss) admissibility basin, without assuming global RG trajectory control.
2. **KP $\Rightarrow$ clustering:** inside the KP basin, a standard cluster expansion yields exponential Euclidean clustering with explicit decay in block distance.
3. **Clustering $\Rightarrow$ spectral gap:** using reflection positivity on the unflowed positive-time algebra A_+ , OS reconstruction converts Euclidean semigroup decay on a dense core into a positive spectral gap.
4. **Physical noncollapse:** a bounded physical entrance scale transports the block decay rate to a uniform physical mass lower bound.

Where group dependence enters Route 1

Route 1 is group-agnostic except at the earliest “mixing/transfer” stage, where one needs uniform control of high-representation tails for the single-plaquette density. This is exactly where **A1** enters: it bounds normalized character coefficients in a crossover form $\exp(-c \cdot C_2/(\beta + \sqrt{C_2}))$. Once A1 is available, the remaining transfer and KP-budget inequalities depend on group only through a finite ledger of named constants.

Block 2: Lane C is universal once A1 is available

Lane C produces a unique continuum state on the flowed observable algebra A_{flow} using a dyadic RG in a polymer seminorm topology with a chart-truncated one-step integration operator (*patched step*). Inputs: uniform finite range and cumulants on a fixed chart event; an extracted-irrelevant scaling gap Δ_{min} ; a one-step expectation bound C_E controlled by moments; and stable-manifold / slaving of irrelevants to the AF coupling trajectory. De-patching then shows the patched RG and the physical Wilson state have identical continuum limits for flowed observables. All of this is identical across P1–P4; group dependence appears only through the constants needed to guarantee the chart regime, ultimately funneled through A1 and a finite constant table.

Block 3: Lane A separates positivity from flow

Lane A performs the sharp local limit $t \downarrow 0$ in a way designed to survive the standard reflection-positivity objection: reflection positivity is used only on the unflowed positive-time algebra A_+ . Flowed probes are replaced inside OS/RP expressions by **RP-safe positive-time representatives** (bounded cylinder functionals supported in $\{x_0 > 0\}$), with an unconditional $O(t^\eta)$ error obtained by quasi-local Gaussian tails plus bad-event absorption deep in the AF window. The small-flow-time expansion (SFTE) is inverted sectorwise; invertibility bounds are proved so that $\Lambda(t)t^\eta \rightarrow 0$. This yields OS axioms and a sharp-local operator algebra without asserting RP for the flowed algebra.

Interlock diagram: what is reused vs what changes

The four papers differ only in the **A1 proof module** and the resulting group-dependent constant ledger entries. All other modules are reused with only notational substitutions.

Reusable core modules (common to all papers)

- RP/Chessboard for Wilson weights in the chosen representation.
- Good-event chart control: plaquette tail bounds $\rightarrow$ chart event $\rightarrow$ uniform convexity $\rightarrow$ Brascamp–Lieb/LSI $\rightarrow$ cumulants.
- Dyadic RG Taylor bounds: analytic one-step map with explicit cubic remainders; irrelevant contraction $\kappa < 1$.
- C3-U packaging: stable manifold + injective tuning + Lipschitz observable dependence $\rightarrow$ no-subsequence convergence.
- Lane A OS closure: RP-safe representatives + bad-event absorption + SFTE + sectorwise invertibility $\rightarrow$ OS axioms.
- Route 1 intake: finite-depth entrance $\rightarrow$ KP clustering $\rightarrow$ OS gap; bounded physical entrance scale $\rightarrow$ physical mass.

Family-specific A1 modules (the only substantive variation)

- **P1 (SU(2))**: rank-one base case; simplest A1 template.
- **P2 (SU(N))**: Toeplitz/TN machinery + Amos/arsinh inequalities.
- **P3 (Spin/Sp)**: Weyl character formulas for types B/C/D + TN moment-matrix tools.
- **P4 (Exceptional)**: maximal-torus parametrization and exceptional root data; determinantal-window plus contour-shift control; certificate protocol for extracting $(c_{A1}(G), \beta_{A1}(G))$.

How referees should verify the series efficiently

A minimal verification plan:

1. Pick one base paper (recommended P1 or P2) and verify the shared architecture once (Lane C, Lane A, Route 1, OS intake).
2. For each remaining paper, verify only (i) the A1 appendix and (ii) the constant-ledger points where A1 feeds the entrance/transfer inequalities.
3. Spot-check scope controls: global-form and line-operator exclusions match the adjoint-local net claim.

Notation alignment across the series

- **A1** always denotes the one-plaquette crossover bound in the canonical form $\exp(-c \cdot C_2/(\beta + \sqrt{C_2}))$.
- q^* denotes the entrance-scale Dobrushin contraction constant; $m_{\text{blk}} = -\log q^*$ is the block decay rate.
- C_{ent} denotes the bounded physical entrance scale constant $a(\beta)L_{\text{ent}}(\beta) \leq C_{\text{ent}}\ell_0$.
- **Patched** refers only to chart-truncated analytic surrogates; final continuum statements are for the unpatched Wilson state via de-patching.